

Bitcoin Difficulty Calculation (Lecture Notes)

Bitcoin difficulty measures how hard it is to mine a block. It is defined as the ratio between a reference target (difficulty-1 target) and the current target.

$$\text{difficulty} = \text{difficulty_1_target} / \text{current_target}$$

1. Understanding nBits Format

Each target is stored in compact form (nBits):

$$\text{nBits} = [\text{exponent}][\text{mantissa}]$$
$$\text{target} = \text{mantissa} \times 2^{(8 \times (\text{exponent} - 3))}$$

2. Step-by-Step Calculation

Difficulty-1 Target:

nBits = 0x1d00ffff

exponent = 29

mantissa = 65535

Current Target:

nBits = 0x1705c739

exponent = 23

mantissa ≈ 378681

3. Simplified Formula

$$\text{difficulty} = (\text{mantissa}_{\text{diff-1}} / \text{mantissa}_{\text{current}}) \times 2^{(8 \times (\text{exponent}_{\text{diff-1}} - \text{exponent}_{\text{current}}))}$$

4. Example Calculation

$$\begin{aligned} \text{difficulty} &= (65535 / 378681) \times 2^{(8 \times (29 - 23))} \\ &= 0.173 \times 2^{48} \\ &\approx 0.173 \times 2.81 \times 10^{14} \\ &\approx 4.9 \times 10^{13} \end{aligned}$$

5. Interpretation

- Higher difficulty → smaller target
- Smaller target → more leading zeros
- More zeros → more hash computations required

Summary

Difficulty is computed using a ratio of targets. Use mantissa and exponent to avoid large numbers. The formula simplifies computation efficiently.